Software Requirements Specification

Version: 2.0 Status: Updated Draft

Document purpose: This SRS defines the functional, non-functional, interface, security and domain requirements for the Hostel Management System.

Revision History

Version	Date	Change Summary	Status
1.0	Earlier version	Initial requirement specification with 11 functional requirements and 9 non-functional requirements.	Superseded
2.0	11-11-2026	Expanded structure, added role-aware requirements, security requirements, acceptance criteria and traceability.	Updated draft

Approvals

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1. Introduction

1.1 Purpose

The purpose of the Hostel Management System is to provide a centralized web-based platform for managing student hostel services and administrative operations. This document specifies what the system shall do and the quality, interface, security and operational constraints it shall satisfy.

1.2 Scope

The system supports student registration and authentication, student profiles, room information and allocation records, fee records and payment history, common facility booking, mess services, leave applications, laundry requests, complaints, lost and found records, and notifications. Warden functions include controlled access to administrative dashboards and management of supported hostel records such as fee records and mess menus.

The system does not define the internal operation of external banking/payment providers, institutional identity systems, SMS/email provider infrastructure, or physical hostel hardware. External integrations are dependencies and must be configured separately.

1.3 Audience

- Students using hostel services.
- Wardens or authorized hostel administrators managing supported records.
- Developers maintaining the Flask application and database.
- Testers validating functional and non-functional requirements.
- Project guide and academic evaluators reviewing the system.

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
HMS	Hostel Management System
SRS	Software Requirements Specification
SAD	Software Architecture and Design Specification
STP	Software Test Plan
FR	Functional Requirement
NFR	Non-Functional Requirement
SR	Security Requirement
RBAC	Role-Based Access Control
ORM	Object-Relational Mapping
UI	User Interface
HTTPS	Hypertext Transfer Protocol Secure

1.5 References

- Hostel Management System requirements and earlier requirement specification supplied for this project.
- Software Engineering Unit 3 lecture material: requirements engineering and architecture/design topics.
- Software Engineering Unit 4 lecture material: software quality, metrics and testing.
- Project source repository and automated test suite, including Flask application and pytest tests.

2. Overall Description

2.1 Product Perspective

HMS is a browser-accessible web application implemented using Python and Flask. The application layer handles requests, validation, authentication and business operations. SQLAlchemy provides object-relational database access. The application is configured to use a relational database in deployment and a separate SQLite database for automated tests. The system is organized into student-facing and warden-facing interfaces with role-restricted operations.

2.2 Major Product Functions

- Student registration, login, logout and profile management.

- Room details, room availability and hostel/room information.
- Fee details, payment processing workflow, payment history and receipts.
- Common facility availability and booking.
- Leave application and status/history tracking.
- Mess menu, meal registration, attendance, feedback, holidays and mess-fee information.
- Laundry requests and status information.
- Complaints with category, description, priority, attachments and status tracking.
- Lost and found item reporting and search/claim workflow.
- Notifications and announcements.
- Warden authentication, dashboard access, fee-record operations and mess-menu administration.

2.3 User Roles and Characteristics

Role	Characteristics and permitted activities
Student	Uses a browser; registers/logs in; accesses own profile and hostel services; views own records; submits bookings, leave requests, complaints and service requests.
Warden	Authorized administrative user; accesses warden dashboard and supported management operations; views or manages relevant hostel records.
Developer / Maintainer	Configures the application, database and environment; maintains code and deployment; not an ordinary end user.

2.4 Operating Environment

- Client: modern desktop or mobile web browser.
- Server: Python runtime with Flask application.
- Persistence: relational database accessed through SQLAlchemy ORM.
- Development and testing: Python virtual environment, pytest and Flask test client.
- The application is developed and tested on a macOS/Linux-compatible Python environment.

2.5 Constraints

- Users require network access and a supported browser.
- Access to protected functions requires successful authentication and an authorized role.
- Database connectivity and valid configuration are required for persistent operations.
- Uploaded identity proof or complaint attachments must conform to configured upload restrictions.
- Payment processing depends on the configured payment workflow/provider; automated tests may use simulated payment behaviour.
- Hostel rules, room capacity and fee policies are controlled by institutional data and configuration.

2.6 Assumptions and Dependencies

- Students provide accurate registration, contact and parent details.
- Each student is associated with a unique institutional identifier.
- Wardens are provisioned with valid credentials and assigned administrative access.
- The database service and application hosting environment are available.
- Payment, email/SMS and other third-party services are external dependencies where enabled.

3. External Interface Requirements

3.1 User Interface

The interface shall provide registration, login, dashboard, profile, room details, fees/payments, facility booking, leave, mess, laundry, lost and found, complaints and notifications views. Warden users shall receive a separate role-appropriate dashboard and management views. Forms shall identify required fields, validate user input and present understandable success or error messages.

3.2 Hardware Interface

No specialized hardware interface is required. Users access the application through desktop, laptop, tablet or smartphone devices. File upload capability depends on the client device and browser.

3.3 Software Interface

- Web browser communicates with the Flask application over HTTP/HTTPS depending on deployment configuration.
- Flask application uses SQLAlchemy to communicate with the configured relational database.
- The payment workflow may connect to an external payment provider when configured; test execution can use a simulated payment flow.
- File upload handling uses application-managed storage.

3.4 Communication Interface

The deployed system should use HTTPS for browser-to-server communication. Database connections should use protected configuration and should not expose credentials in source control. External provider communication, if enabled, must use the provider-supported secure protocol.

4. System Features and Functional Requirements

Each requirement includes a priority and acceptance criteria. Test references are logical test identifiers; they are used in the RTM and Test Plan.

Req ID	Feature	Priority	Requirement	Acceptance Criteria	Test Reference
FR-001	Student Registration	High	The system shall allow a student to register using required identity, personal, contact, academic, parent and hostel information, including required proof upload where configured.	A valid submission creates one student account; invalid or incomplete submissions are rejected with field-level feedback.	TC-REG-01, TC-REG-02
FR-002	Student Authentication and Session	High	The system shall authenticate students and wardens and	Valid credentials reach the correct role dashboard; invalid credentials are rejected; logout ends	TC-AUTH-01 to TC-AUTH-04

			shall provide logout and protected-session behaviour.	access to protected pages.	
FR-003	Student Profile	Medium	The system shall allow students to view and update permitted profile information and view associated room, fee, leave, complaint and booking records.	A student can access their own profile and related records; unauthorized records are not disclosed.	TC-PROF-01, TC-SEC-01
FR-004	Room Management and Details	High	The system shall provide room information, capacity, availability, hostel filtering, allocation details and maintenance status where maintained.	Room information is displayed consistently; unavailable or invalid room data is not presented as available.	TC-ROOM-01 to TC-ROOM-04
FR-005	Common Facility Booking	Medium	The system shall allow students to view facility availability, select a valid date/time slot, book or cancel a booking and view booking history.	A valid available slot can be booked; unavailable/conflicting slots are rejected; booking history reflects the result.	TC-FAC-01 to TC-FAC-03
FR-006	Fee Management and Payment	High	The system shall display fee details and due information, process the configured payment workflow, and maintain payment status, history and receipt information.	A successful payment workflow creates a payment record and updates the relevant status; invalid or unauthorized payment attempts are rejected.	TC-FEE-01 to TC-FEE-04

FR-007	Mess Management	Medium	The system shall provide mess menu and meal-related information, including meal registration, attendance, feedback, holidays and mess-fee information where enabled.	Students can access applicable mess information and submit permitted meal-related actions; records are stored consistently.	TC-MESS-01 to TC-MESS-04
FR-008	Leave Management	High	The system shall allow a student to submit a leave application with leave type, dates/duration, reason and parent contact details, and view its status/history.	A valid application is recorded and associated with its student; invalid details are rejected; status/history is visible to the owner.	TC-LEAVE-01 to TC-LEAVE-03
FR-009	Laundry Management	Medium	The system shall support laundry request submission and display item, pickup/delivery, charge, payment and request-status information where the module is enabled.	A valid request is recorded and its status/history can be retrieved by the associated student.	TC-LAUN-01
FR-010	Lost and Found	Low	The system shall allow users to report lost/found items with details and location, search available reports and record claim/return information where enabled.	Submitted item details can be retrieved and the claim/return status is reflected.	TC-LF-01
FR-011	Complaint Management	High	The system shall allow students to	A valid complaint is stored under the submitting student;	TC-COMP-01 to TC-COMP-03

			submit categorized complaints with description, priority and optional attachment, and track complaint status.	attachment access is controlled; status can be tracked.	
FR-012	Warden Role and Dashboard	High	The system shall provide a warden login flow and restrict warden dashboard and administrative functions to authorized warden accounts.	A valid warden reaches the warden dashboard; students cannot access warden-only pages; unauthorized requests are denied.	TC-WARD-01 to TC-WARD-03
FR-013	Warden Fee Operations	High	The system shall allow an authorized warden to view relevant fee records and create or manage supported fee records.	Authorized operations persist valid fee data; invalid entries are rejected; student access remains limited to own fee records.	TC-WFEE-01, TC-SEC-02
FR-014	Warden Mess Menu Administration	Medium	The system shall allow an authorized warden to create, update and delete mess menu entries and reject duplicate or invalid entries.	Valid changes are persisted and reflected in the menu; invalid/duplicate entries are rejected.	TC-WMENU-01 to TC-WMENU-03
FR-015	Notifications and Announcements	Medium	The system shall provide student-facing notices and notifications for relevant hostel information where configured.	Available notices can be displayed to intended users; read/unread or status information is retained where implemented.	TC-NOT-01

5. Non-Functional Requirements

Req ID	Quality Attribute	Priority	Requirement	Acceptance / Measurement
NFR-001	Performance	High	The system shall respond to common page and form requests within an acceptable time under normal expected usage.	Record response time for representative requests; investigate slow requests. No measured performance claim is made without a load test.
NFR-002	Usability	High	The system shall provide consistent navigation, readable content, clear validation feedback and responsive layouts.	Representative student and warden workflows can be completed without ambiguous form feedback.
NFR-003	Security	High	The system shall authenticate users, protect passwords, enforce role-based access and prevent unauthorized access to protected resources.	Authentication and access-control tests pass; credentials are not stored as plaintext.
NFR-004	Reliability	High	The system shall handle invalid requests and database errors without corrupting existing records.	Validation and transaction tests demonstrate correct failure handling and record consistency.
NFR-005	Availability	Medium	The system shall be deployable in a stable hosting environment with documented recovery and maintenance procedures.	Deployment and service recovery procedures are documented; uptime target must be set by deployment owner.
NFR-006	Scalability	Medium	The system design shall permit growth in student records and concurrent usage without requiring redesign of core modules.	Database and application configuration can be scaled; capacity requires load testing before a numeric limit is claimed.
NFR-007	Maintainability	Medium	The application shall use modular	Code review and automated test

			routes, models and templates with readable code and automated tests.	suite support maintenance; regression suite can be executed.
NFR-008	Data Integrity	High	The system shall maintain accurate student, room, fee, booking and service records and preserve relationships between records.	Database constraints and functional tests prevent invalid or unauthorized record changes.
NFR-009	Compatibility and Portability	Medium	The system shall support commonly used modern browsers and deployment on compatible Python/server environments.	Smoke tests are performed on target browsers and deployment environment before release.

6. Security Objectives and Requirements

6.1 Security Objectives

1. Confidentiality: protect personal, academic, parent-contact, fee and complaint information from unauthorized disclosure.
2. Access control and accountability: ensure students and wardens can access only the functions and records permitted for their roles, with relevant operations attributable to authenticated sessions.

6.2 Security Requirements

Security ID	Requirement	Priority	Acceptance Criteria
SR-001	The system shall require authentication before granting access to protected student or warden functions.	High	Protected route access without a valid session is denied.
SR-002	The system shall enforce role-based authorization for warden-only operations.	High	A student session cannot access warden dashboard or management operations.
SR-003	The system shall store passwords using a suitable one-way password-hashing mechanism and shall never store plaintext passwords.	High	Stored password values are hashes and authentication verifies the hash.
SR-004	The system shall validate and constrain user-supplied form fields and uploaded files before	High	Invalid values and disallowed uploads are rejected safely.

	processing or storing them.		
SR-005	The deployed system shall use HTTPS and protect database credentials and secret configuration outside publicly accessible source code.	High	Deployment configuration enforces TLS and secrets are supplied through protected configuration.
SR-006	The system shall restrict access to student-specific records so a student can view or modify only records belonging to their account.	High	Cross-student access attempts are denied.

7. Quality Attributes and Acceptance Tests

The quality model used for evaluation considers functionality, reliability, usability, efficiency, maintainability and portability. These attributes follow the quality-oriented discussion in the Unit 4 material. Numeric performance, availability and scalability targets are not asserted until measured in the target environment.

Quality Attribute	**HMS Interpretation**	**Acceptance Test / Evidence**
Functionality	Required hostel workflows behave correctly.	Execute requirement-linked functional test cases.
Reliability	Operations preserve data consistency and handle errors.	Invalid input, database interaction and regression tests.
Usability	Students and wardens can navigate their respective tasks.	Task walkthrough and interface review.
Efficiency	Pages and transactions respond in a reasonable time.	Measure response times under representative and load conditions.
Maintainability	Modules can be corrected and extended without destabilizing unrelated features.	Code review, modularity review and regression suite.
Portability	Application runs in supported browser and server environments.	Deployment smoke tests across target environments.

8. UML Use-Case Models

Use-case model 1: Student interactions — Register, Login, View Profile, View Room Details, View Fees, Make Payment, Book Facility, Apply Leave, Access Mess, Submit Laundry Request, Report Lost/Found, Submit Complaint and View Notifications.

Use-case model 2: Warden interactions — Warden Login, View Warden Dashboard, View Fee Records, Create/Manage Fee Records, Manage Mess Menu and access authorized hostel administration functions.

The diagrams should show Student and Warden as separate actors and connect each actor only to its permitted use cases. Include/extend relationships should be used only where one use case is necessarily included in or optionally extends another.

9. Requirements Traceability Matrix

Requirement ID	Requirement / Module	Test Case Reference	Verification Status
FR-001	Student Registration	TC-REG-01, TC-REG-02	Covered by functional test design
FR-002	Student Authentication and Session	TC-AUTH-01 to TC-AUTH-04	Covered by functional test design
FR-003	Student Profile	TC-PROF-01, TC-SEC-01	Covered by functional test design
FR-004	Room Management and Details	TC-ROOM-01 to TC-ROOM-04	Covered by functional test design
FR-005	Common Facility Booking	TC-FAC-01 to TC-FAC-03	Covered by functional test design
FR-006	Fee Management and Payment	TC-FEE-01 to TC-FEE-04	Covered by functional test design
FR-007	Mess Management	TC-MESS-01 to TC-MESS-04	Covered by functional test design
FR-008	Leave Management	TC-LEAVE-01 to TC-LEAVE-03	Covered by functional test design
FR-009	Laundry Management	TC-LAUN-01	Covered by functional test design
FR-010	Lost and Found	TC-LF-01	Covered by functional test design
FR-011	Complaint Management	TC-COMP-01 to TC-COMP-03	Covered by functional test design
FR-012	Warden Role and Dashboard	TC-WARD-01 to TC-WARD-03	Covered by functional test design
FR-013	Warden Fee Operations	TC-WFEE-01, TC-SEC-02	Covered by functional test design
FR-014	Warden Mess Menu Administration	TC-WMENU-01 to TC-WMENU-03	Covered by functional test design
FR-015	Notifications and Announcements	TC-NOT-01	Covered by functional test design
NFR-001	Performance	NFR test / measurement	Planned / requires measurement
NFR-002	Usability	NFR test / measurement	Planned / requires measurement
NFR-003	Security	NFR test / measurement	Planned / requires measurement
NFR-004	Reliability	NFR test / measurement	Planned / requires measurement
NFR-005	Availability	NFR test / measurement	Planned / requires measurement
NFR-006	Scalability	NFR test / measurement	Planned / requires measurement
NFR-007	Maintainability	NFR test / measurement	Planned / requires measurement

NFR-008	Data Integrity	NFR test / measurement	Planned / requires measurement
NFR-009	Compatibility and Portability	NFR test / measurement	Planned / requires measurement
SR-001	Security	Security test / review	Planned / partly automated
SR-002	Security	Security test / review	Planned / partly automated
SR-003	Security	Security test / review	Planned / partly automated
SR-004	Security	Security test / review	Planned / partly automated
SR-005	Security	Security test / review	Planned / partly automated
SR-006	Security	Security test / review	Planned / partly automated

10. Appendices

10.1 Major Modules

- Student Registration
- Student Authentication and Session
- Student Profile
- Room Management and Details
- Common Facility Booking
- Fee Management and Payment
- Mess Management
- Leave Management
- Laundry Management
- Lost and Found
- Complaint Management
- Warden Role and Dashboard
- Warden Fee Operations
- Warden Mess Menu Administration
- Notifications and Announcements

10.2 Implementation and Verification Note

The project automated test suite was previously reported as 223 passed, 0 failed and 7 warnings. This is a test-suite result, not proof that every acceptance criterion, browser, external integration, performance target or manual workflow has been independently verified. Features described as “where enabled” or dependent on external providers require confirmation against the deployed configuration.

11. Requirements Engineering Notes and Change Control

The requirements baseline is organized so that each requirement is uniquely identified, prioritized, stated in verifiable language and linked to one or more verification references. Each approved change shall be recorded with its reason, affected requirement IDs, impact on scope, design and testing, approval decision and document version.

- Forward traceability follows each requirement from the SRS to its architecture component, design/interface element and test case.
- Backward traceability follows each design or test item to the requirement that justifies it, helping identify unnecessary or unapproved implementation.
- Approved requirement changes shall be reflected in the Requirements Traceability Matrix (RTM), Software Architecture and Design Specification (SAD), and Software Test Plan (STP).
- The SRS remains the baseline for design and testing; unresolved or unapproved changes shall not silently alter the requirements baseline.

12. Requirement Verification Conventions

Functional requirements shall be verified through observable outcomes such as account creation, access denial, record persistence, booking creation or status display. Non-functional requirements shall be verified through review or measurement. Where a numerical target has not been agreed or measured, the document records the measurement method rather than claiming an unsupported threshold.

The functional, non-functional and security requirements in Sections 4–6 are the normative requirement set. Section 9 provides the mapping to test references. Requirements marked planned or partly automated require additional execution evidence before acceptance.